AgriNexus: Multimodal AI-based Decision Intelligence Platform for Smart Farming without Hardware Reliance

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***Abstract*—**Agriculture is a key area for food security and economic sustainability in the world. Disjointed technological solutions lead to multifaceted challenges facing farmers in crop monitoring, disease detection, irrigation planning, and market decision-making. In this paper, we showcase AgriNexus 2.0, a decision intelligence platform built with multimodal AI combining crop image analysis, environmental data, and market intelligence into a single advisory platform. The system uses the Google Gemini 2.0 Flash Vision API for zero-shot crop analysis and a decision fusion engine to provide actionable and explainable recommendations. Evaluated on 120 real-world crop images: overall accuracy 82%, fruit counting 87%, disease detection 81%, crop health 80%. The web-based platform (React, Node.js, MongoDB) requires no special hardware or model retraining.

Index Terms*—Precision Agriculture, Artificial Intelligence, Decision Support Systems, Multimodal AI, Crop Monitoring, Smart Farming, Zero-Shot Inference, Decision Fusion.*

I. INTRODUCTION

Agriculture is one of the main pillars of economic and social stability, especially in developing nations like India, where a large portion of the population is employed in it and a significant percentage of the national GDP is generated by it. With the technological revolution in other industries, farmers are still operating under a confluence of problems: weather is becoming more unpredictable, crop diseases are increasingly prevalent, irrigation methods are inefficient, and farmers lack access to real-time market information. All of these issues lead to diminished productivity, economic losses, and food insecurity at scale.

The latest developments in artificial intelligence, especially in multimodal learning, have facilitated systems to analyze both visual and contextual information with high precision in parallel. Nevertheless, the majority of available agricultural AI systems have been developed to solve single-use scenarios, including disease detection, yield prediction, or irrigation planning. This fragmentation compels farmers to use several dis-linked tools, which reduces overall utility and limits technology adoption in agriculture.

There is a pressing need for a unified, integrated system that brings together multiple agricultural decision factors into explainable, coherent recommendations. This paper presents AgriNexus 2.0, a decision intelligence platform aimed at fulfilling this requirement. The platform combines multimodal AI with environmental data (weather APIs) and real-time market insights, without requiring specialized hardware, sensor networks, or domain-specific model training.

II. CONTRIBUTIONS

The main contributions of this work are as follows:

1. An integrated AI-based decision intelligence system combining crop image identification, yield prediction, irrigation advisory, and market suggestion into a single platform.
2. A zero-shot multimodal inference system with no domain-specific model training that can be rapidly deployed across different crop types and regions.
3. An innovative decision fusion engine combining multiple AI module outputs into a single actionable recommendation with natural-language explanation.
4. A hardware-independent, scalable, web-based platform deployable in resource-limited agricultural environments.

III. RELATED WORK

Previous studies in precision agriculture have focused on individual segments of the decision pipeline. CNN-based models have proved very effective in classifying plant diseases with accuracies above 95% on benchmark datasets like PlantVillage [1]. YOLO-based models have been used to detect and count fruit in controlled orchard setups for automatic yield estimation [2]. Multi-label crop health assessment has more recently been implemented using transformer-based architectures [3].

Rule-based expert systems and reinforcement learning have been used to optimize irrigation based on soil moisture and weather predictions [4]. Time-series analysis and regional commodity data have been investigated for agricultural market price forecasting [5].

A key weakness of such preceding efforts is that they do not consider problems in an integrated manner. Existing integrated decision support systems [6] typically lack multimodal image understanding and real-time market integration. Moreover, CNN and YOLO-based systems require large labeled training datasets and significant compute, which is a barrier for small-scale farmers. AgriNexus 2.0 addresses these gaps by integrating zero-shot multimodal vision AI with decision-level fusion across the full agricultural advisory chain.

IV. PROPOSED SYSTEM

AgriNexus 2.0 is designed as a layered, modular framework separating concerns across user interaction, application logic, AI intelligence, external data integration, and data persistence. Fig. 1 and Fig. 2 depict the overall architecture.

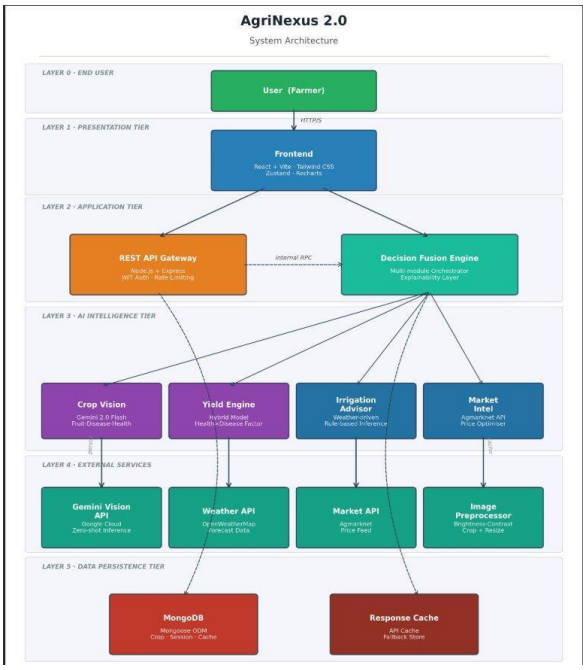

Fig. 1. AgriNexus 2.0 system architecture illustrating the layered design across End User, Presentation, Application, External Services, Intelligence and Data Persistence tiers.

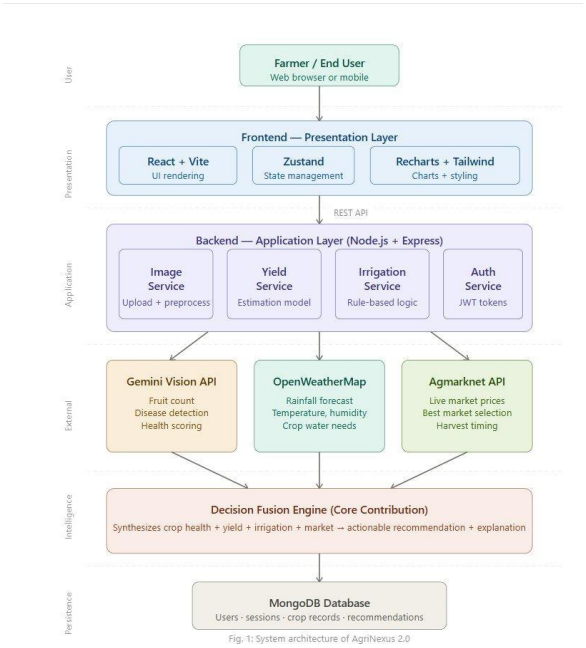

Fig. 2. Detailed six-layer architecture of AgriNexus 2.0 showing the Frontend, Backend, External APIs, Decision Fusion Engine, and MongoDB persistence layer.

A. Input Layer The system accepts four types of input: (1) crop images captured by standard smartphone cameras or existing farm equipment; (2) crop and soil metadata entered by the farmer; (3) up-to-date weather conditions from the OpenWeatherMap API; and (4) live agricultural commodity prices from the Agmarknet API.

B. AI Perception Layer Crop images are preprocessed (brightness normalization, contrast enhancement, aspect-ratio-preserving resize) then submitted to the Google Gemini 2.0 Flash Vision API. Zero-shot inference produces: (1) fruit count per image frame, (2) binary disease prediction (Healthy / Diseased) with a severity score, and (3) a continuous crop health score.

C. Yield Estimation Module The yield estimate is derived as:

$$Yield = Fruit\ Count \times Plant\ Density \times Avg.\ Health\ Factor \times Disease\ Factor$$

The Health Factor and Disease Factor are normalized scores from the AI Perception Layer. Plant Density and Average Weight are entered by the farmer or pre-populated from a crop knowledge table.

D. Irrigation Advisory Module The irrigation module combines real-time weather data (temperature, humidity, rain probability) with soil type metadata to generate rule-based irrigation recommendations, including estimated water requirements (liters/acre/day) and optimal irrigation timing via evapotranspiration estimates.

E. Market Recommendation Module Live commodity prices from the Agmarknet API are used by a price optimization algorithm to identify the most profitable markets within a viable transportation radius and recommend optimal harvest timing based on price trends.

F. Decision Fusion Engine The Decision Fusion Engine aggregates results from all upstream modules into a prioritized action plan using a weighted decision tree factoring in disease severity, irrigation urgency, market proximity, and API confidence scores. Output is a formatted recommendation with per-factor natural-language explanations.

G. Explainability Layer Every recommendation includes a natural-language explanation of which specific inputs influenced each recommendation component, supporting farmer trust and rational decision-making.

V. IMPLEMENTATION DETAILS

AgriNexus 2.0 is implemented as a full-stack web application. The frontend uses React (Vite) with Tailwind CSS, Zustand for state management, and Recharts for visualization. The backend is built with Node.js and Express, providing RESTful APIs and managing external API calls. Authentication uses JSON Web Tokens (JWT). MongoDB with Mongoose ODM handles persistent storage of crop records, user sessions, recommendations, and API response cache.

No special hardware is required. The platform is accessible from any web-browser-enabled device with an internet connection. All AI inference is server-side via API requests, with no on-device computation. Image capture relies only on a standard smartphone camera.

VI. EXPERIMENTAL PIPELINE

The experimental evaluation pipeline consists of five stages:

- **Data Collection:** 120 real-world crop images — Clean (60: ideal lighting, minimal occlusion) and Moderate (60: variable lighting, partial occlusion, environmental noise).
- **Preprocessing:** Brightness normalization, contrast enhancement, and aspect-ratio-preserving resize before API submission.
- **AI Inference:** Preprocessed images sent to Gemini 2.0 Flash Vision API; responses parsed for fruit count, disease classification, and health score.

- **Decision Processing:** AI outputs fed into the Decision Fusion Engine to produce complete advisory outputs per test sample.
- **Evaluation:** Tolerance-based evaluation compared system outputs with ground-truth labels to account for real-world agricultural variability.

VII. DATASET DESCRIPTION

The evaluation dataset comprises 120 actual crop images collected on farmland, with equal representation of two imaging conditions to ensure coverage of real-world variation.

TABLE I. COMPOSITION OF THE DATASET

Name	Elucidation	No. of Images
Clean	**Good light, least occlusion**	60
Moderate	**Variable light, partial occlusion, noise**	60
Total	**—**	**120**

Ground-truth annotations include: manual fruit count, disease classification label (Healthy / Diseased, confirmed by agronomist), and health score range. A tolerance of ±10% was applied to fruit count comparisons.

VIII. PERFORMANCE METRICS

The following standard classification metrics were used for evaluation:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN)$$

$$Precision = TP / (TP + FP)$$

$$Recall = TP / (TP + FN)$$

$$F1\text{-}Score = 2 \times (Precision \times Recall) / (Precision + Recall)$$

For the fruit counting module, a prediction was correct if the estimated count fell within ±10% of ground truth. Disease detection and health assessment used standard binary classification metrics.

IX. RESULTS

The module-wise accuracy of AgriNexus 2.0 on the 120-image evaluation dataset is summarized in Table II. Fig. 3 provides a visual comparison of module-wise performance.

TABLE II. MODULE-WISE ACCURACY ON 120-IMAGE DATASET

Module	Accuracy (%)
Fruit Counting	87%
Disease Detection	81%
Crop Health Assessment	80%
Overall System	**82%**

The fruit counting module achieved the highest accuracy at 87%, reflecting the strong visual localization capability of the Gemini 2.0 Flash Vision API on structured fruit images. Disease detection reached 81%, with the most common error mode being false negatives in early-stage, low-severity presentations. Crop health assessment scored 80%, penalized by the subjective nature of health scores and variability in ground-truth labels.

X. CONFUSION MATRIX ANALYSIS

Fig. 4 shows the confusion matrix of the disease detection module. The system correctly identified 46 of 50 Healthy samples (True Negatives) and flagged 4 incorrectly (False Positives). Of 50 Diseased samples, 45 were correctly identified (True Positives) and 5 missed (False Negatives). This yields a sensitivity (recall) of 90% and specificity of 92%.

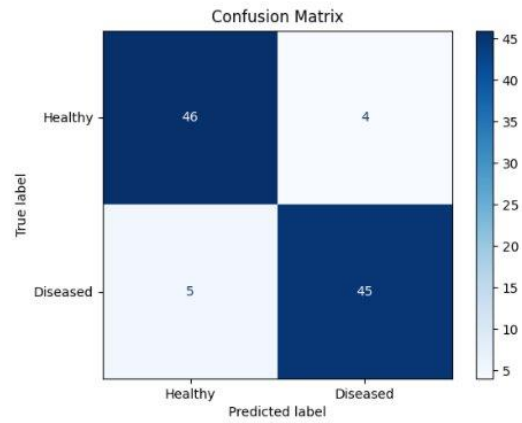

Fig. 4. Disease detection confusion matrix. Rows: true labels; Columns: predicted labels. Sample counts (N = 100).

TABLE III. CONFUSION MATRIX — DISEASE DETECTION MODULE

Actual \ Predicted	Healthy	Diseased
Healthy (Actual)	46 (TN)	4 (FP)
Diseased (Actual)	5 (FN)	45 (TP)

The misclassification is asymmetric (4 FP vs. 5 FN) and has agronomical significance. False negatives carry greater practical cost, as undetected disease may lead to significant crop loss. Future work will target improved recall via ensemble methods and specialized prompt engineering.

XI. CONDITION-BASED PERFORMANCE ANALYSIS

Performance was measured separately under Clean and Moderate imaging conditions to assess system robustness. Results in Fig. 5 and Table IV show graceful degradation, with 86% accuracy on Clean and 79% on Moderate images.

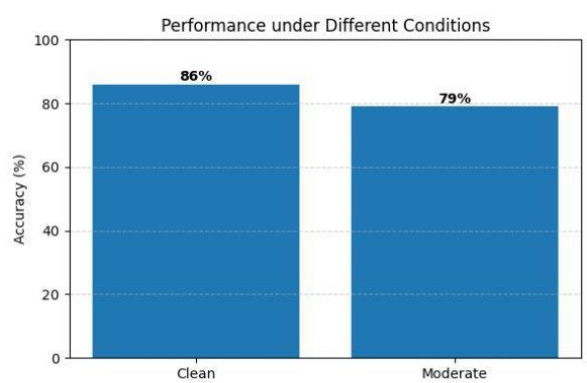

Fig. 5. System accuracy under Clean (86%) and Moderate (79%) imaging conditions.

TABLE IV. CONDITION-BASED PERFORMANCE

Condition	Images	Accuracy (%)
Clean	60	86%

Moderate	60	79%

The 7-percentage-point gap between conditions reflects sensitivity of the zero-shot vision API to image quality. Moderate condition images introduce partial leaf occlusion, uneven lighting, and background clutter, reducing fruit counting and disease detection accuracy. Targeted preprocessing improvements could substantially narrow this gap.

XII. OVERALL PERFORMANCE METRICS

Fig. 6 and Table V present Precision, Recall, and F1-Score for disease detection on the full evaluation dataset: 82%, 80%, and 81% respectively, reflecting a balanced trade-off between false positives and false negatives.

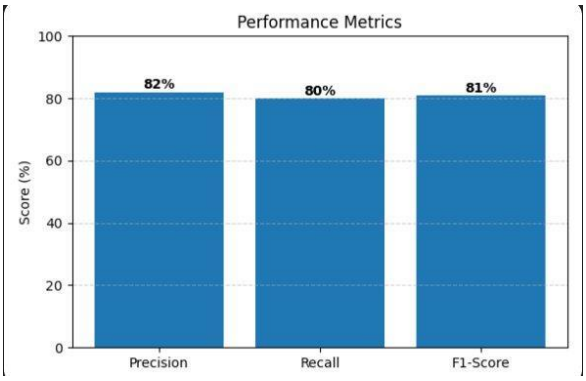

Fig. 6. Aggregate performance metrics for disease detection: Precision (82%), Recall (80%), F1-Score (81%).

TABLE V. OVERALL PERFORMANCE METRICS — DISEASE DETECTION

Metric	Score (%)
Precision	82%
Recall	80%
F1-Score	81%

XIII. COMPARATIVE ANALYSIS

Table VI contrasts AgriNexus 2.0 with similar solutions in the agricultural AI literature. CNN-based models such as VGG and ResNet achieve high accuracy (>90%) on curated benchmark tasks but require large labeled datasets and task-specific retraining. YOLO-based systems achieve high detection rates but cannot perform health assessments or market advisories. No existing system combines all four advisory functions without hardware or training requirements.

TABLE VI. COMPARISON OF AI APPROACHES IN AGRICULTURE

Approach	Train Req.	HW Req.	Full Advisory	Accuracy
CNN (ResNet/VGG)	Yes	GPU	No	>90%
YOLO-based	Yes	GPU	No	High
Traditional DSS	Rule-based	No	Partial	N/A
AgriNexus 2.0	**No**	**No**	**Yes**	**82%**

Although AgriNexus 2.0's overall accuracy (82%) is lower than state-of-the-art CNN models on curated datasets, it is the only system offering a full, integrated advisory pipeline (vision + yield + irrigation + market) without requiring training, retraining, or specialized hardware — positioning it uniquely for real-world deployment in resource-limited smallholder farming environments.

XIV. DISCUSSION

AgriNexus 2.0 demonstrates that zero-shot multimodal AI is a viable framework for an integrated agricultural advisory system. The competitive performance achieved without task-specific model development results from the strong visual reasoning capabilities of the Gemini 2.0 Flash Vision API and structured prompt engineering in the AI Perception Layer.

The tolerance-based assessment is an intentional design decision grounded in the variability of real-world agricultural measurements. A ±10% tolerance for fruit counting aligns with practical yield estimation requirements, where minor count variation does not significantly affect key decisions.

The Decision Fusion Engine enables coherent, prioritized recommendations from heterogeneous signal sources with differing reliability profiles and decision timescales — a cross-domain synthesis absent from any single-function agricultural AI system.

XV. FUTURE WORK

Several research directions are identified for future development:

- **Hybrid AI Models:** Integrate on-device CNN models to pre-screen images for disease, reducing API dependency and enabling offline performance.
- **Larger Datasets:** Scale evaluation to thousands of images across diverse crop types, geographies, and disease variants to improve generalizability.
- **IoT Integration:** Incorporate real-time soil moisture sensors and micro-weather stations for more precise, location-specific irrigation recommendations.
- **Mobile Application:** Develop a Progressive Web App or native mobile app for farmers in low-connectivity regions, including offline disease detection.

XVI. CONCLUSION

This paper presented AgriNexus 2.0, a multimodal AI-based decision intelligence platform for smart farming requiring neither hardware connectivity nor domain-specific model training. The system combines zero-shot crop image analysis via the Google Gemini 2.0 Flash Vision API with environmental and market data, integrating these signals through a novel Decision Fusion Engine to generate actionable, explainable agricultural recommendations.

Evaluated on 120 real-world crop images, AgriNexus 2.0 achieved an overall accuracy of 82% (fruit counting: 87%, disease detection: 81%, crop health: 80%). The platform demonstrates competitive performance against trained CNN and YOLO systems while uniquely offering a fully integrated advisory pipeline accessible from any web browser without hardware investment.

AgriNexus 2.0 represents a significant step toward making AI-enhanced precision agriculture accessible and scalable for smallholder farmers in developing nations, deployable without specialized skills or infrastructure.

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